College of Engineering – Department of Civil Engineering

CURRICULUM

Bachelor of Science in Sanitary Engineering

Academic Year 2025-2026

Reference CMOs: CMO No. 98 S 2017, CMO No. 20 s. 2013, CMO No. 39 s. 2021, CMO No. 40 s. 2021

Curriculum Description

The BSSE program recognizes the importance of Sanitary Engineering in the protection and preservation of the environment as well as the promotion of health and well-being of the general public.

Program Education Objectives

The sanitary engineering alumni three to five years after graduation shall:

PEO 1: Specialist
Practice as specialist in solving complex sanitary engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, and economical requirements.

PEO 2: Professionalism and Leadership
Assume leadership position in industry, academe, government, or private sector within consideration to social and ethical responsibility.

PEO 3: Lifelong Learning
Engage in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:

SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex sanitary engineering problems;

SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;

SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.

SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO5 Problem Analysis. Identify, formulate, and solve complex sanitary engineering problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in

engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex sanitary engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex sanitary engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Differential Equations	3	0	3
Probability and Statistics for AI	3	0	3
Numerical Solutions to Engineering Problems	2	3	3
Subtotal	**16**	**3**	**17**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Modern Biology	3	0	3
Physics for Engineers	3	3	4
Geology	3	0	2
Environmental Science and Engineering	3	0	3
Subtotal	**15**	**3**	**16**
C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1
Engineering Drawing	0	3	1
Computer-Aided Design	0	3	1
Engineering Economics	3	0	3
Technopreneurship	3	0	3
Engineering Management	2	0	2
Statics of Rigid Bodies	3	0	3
Dynamics of Rigid Bodies	2	0	2

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Mechanics of Deformable Bodies	4	0	4
Sanitary Engineering Orientation	1	0	1
Computer Programming 1	0	3	1
Computer Programming 2	0	3	1
Subtotal	**18**	**15**	**23**
D. Allied Courses			
Engineering Utilities 1 / Basic Mechanical Engineering	3	0	3
Engineering Utilities 2 / Basic Electrical Engineering	3	0	3
Subtotal	**6**	**0**	**6**
E. Professional Courses			
Basic Occupational Safety and Health	2	0	3
Capstone Design Project 1	1	3	2
Capstone Design Project 2	0	6	2
Construction Materials and Testing	2	3	3
Construction Methods and Project Management	3	3	4
Environmental and Sanitary Chemistry	2	3	3
Environmental Engineering Laboratory	0	3	1
Environmental Planning, Laws and Impact Assessment	3	0	3
Fundamentals of Surveying	3	6	5
Geotechnical Engineering 1 (Soil Mechanics)	3	3	4
GIS Application for Sanitary Engineers	0	3	1
Groundwater and Soil Pollution Remediation	2	0	2
Hydraulics	4	0	4
Hydraulics Laboratory	0	3	1
Hydrology	3	0	3
Methods of Research	3	0	3
Microbiology and Parasitology for Environmental Engineers	2	3	3
Principles of Reinforced/Prestressed Concrete Design	3	3	4
Public Health Engineering	1	3	2
Sanitary Science, Plumbing and Fire Protection as Applied to Buildings	2	3	3
SE Laws, Contracts and Ethics	3	0	3
SE Practice with Comprehensive Examinations	0	6	2
Sewage and Industrial Wastewater Treatment	3	0	3
Sewerage and Urban Drainage	3	0	3
Solid and Hazardous Waste Engineering	3	0	3
Structural Theory	3	3	4
Water Purification Process Design	3	0	3
Water Supply Planning and Development	3	0	3
Subtotal	**60**	**57**	**80**
F. On-the-Job Training			
OJT/Internship (320 hrs)	-	-	**4**
Subtotal	**-**	**-**	**84**
TOTAL TECHNICAL COURSES	**115**	**78**	**146**
II. NON-TECHNICAL COURSES			

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A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modem World	3	0	3
The Contemporary World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Electives and Mandated Courses			
Life and Works of Rizal	3	0	3
Linear Algebra for AI	3	0	3
Machine Learning and Neural Networks	2	3	3
The Entrepreneurial Mind	3	0	3
Subtotal	**11**	**3**	**12**
C. Physical Activities Toward Health and Fitness			
Movement Competency Training	2	0	2
Exercise-based Fitness Activities	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 1	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 2	2	0	2
Subtotal	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Subtotal	**6**	**3**	6
TOTAL NON-TECHNICAL COURSES	**49**	**6**	**50**
GRAND TOTAL	**164**	**84**	**196**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	**17**
B. Natural/Physical Sciences	**16**
C. Basic Engineering Sciences	**23**
D. Allied Courses	**6**
E. Professional Courses	**80**
F. On-the-Job Training	**4**
II. NON-TECHNICAL COURSES	
A. General Education Courses	**24**
B. General Education Electives and Mandated Courses	**12**
C. Physical Activities Toward Health and Fitness	**8**
D. National Service Training Program	**6**
GRAND TOTAL	**196**

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CURRICULUM STRUCTURE

BACHELOR OF SCIENCE IN SANITARY ENGINEERING

Based on the CHED Memorandum Order No. 98 S 2017

FIRST YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CHEM 101	Chemistry for Engineers	4	3	3			TC-B
MATH 101	Differential Calculus	4	4	0			TC-A
ENGG 101	Introduction to Engineering	1	0	3			TC-C
CpE 102	Computer Programming 1	1	0	3			TC-C
GEd 104	The Contemporary World	3	3	0			NTC-A
GEd 105	Readings in Philippine History	3	3	0			NTC-A
GEd 108	Art Appreciation	3	3	0			NTC-A
PATHFit 1	Movement Competency Training	2	2	0			NTC-C
NSTP 111	National Service Training Program 1	3	3	0			NTC-D
	Total	**24**	**21**	**9**			

FIRST YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
MATH 102	Integral Calculus	4	4	0	MATH 101		TC-A
PHYS 101	Physics for Engineers	4	3	3	MATH 101		TC-B
GEd 101	Understanding the Self	3	3	0			NTC-A
GEd 102	Mathematics in the Modern World	3	3	0			NTC-A
GEd 106	Purposive Communication	3	3	0			NTC-A
GEd 109	Science, Technology and Society	3	3	0			NTC-A
CpE 103	Computer Programming 2	1	0	3			TC-C
ENGG 102	Engineering Drawing	1	0	3			TC-C
PATHFit 2	Exercise-based Fitness Activity	2	2	0	PATHFit 1		NTC-C
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111		NTC-D
	Total	**27**	**24**	**9**			

FIRST YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
BioE 101	Modern Biology	3	3	0	PHYS 101		TC-B
GEd 107	Ethics	3	3	0			NTC-B
GEd 103	Life and works of Rizal	3	3	0			NTC-A

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	Total	**9**	**9**	**0**			

SECOND YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
MATH 103	Differential Equations	3	3	0	MATH 102		TC-A
CE 203	Fundamentals of Surveying	5	3	6	ENGG 102		TC-E
SE 201	Sanitary Engineering Orientation	1	1	0			TC-C
SE 202	Environmental and Sanitary Chemistry	3	2	3	CHEM 101		TC-E
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102		TC-C
GeoE 200	Geology	2	2	0			TC-B
ENGG 107	Statics of Rigid Bodies	3	3	0	PHY 101		TC-C
ENGG 108	Dynamics of Rigid Bodies	2	2	0	PHY 101		TC-C
AI 101	Linear Algebra for AI	3	3	0			TC-F
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	2	0	PATHFit 1 & 2		NTC-C
	Total	**25**	**21**	**12**			

SECOND YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
AI 102	Probability and Statistics for AI	3	3	0			TC-A
ENGG 111	Mechanics of Deformable Bodies	4	4	0	ENGG 107		TC-C
SE 203	Numerical Solutions to Engineering Problems	3	2	3	MATH 103		TC-A
SE 204	Microbiology and Parasitology for Environmental Engineers	3	2	3	SE 202		TC-E
ENGG 113	Environmental Science and Engineering	3	3	0			TC-B
EE 209	Engineering Utilities 1	3	3	0			TC-D
ME 209	Engineering Utilities 2	3	3	0			TC-D
CE 308	Hydrology	3	3	0			TC-E
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	2	0	PATHFit 1 & 2		NTC-C
	Total	**27**	**25**	**6**			

THIRD YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 106	Engineering Management	2	2	0	3rd Year Standing		TC-C

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CE 301	Structural Theory	4	3	3	ENGG 111		TC-E
CE 205	Hydraulics	4	4	0	ENGG 111, ENGG 108		TC-E
CE 302	Hydraulics Laboratory	1	0	3	ENGG 111, ENGG 108		TC-E
CE 306	Geotechnical Engineering 1 (Soil Mechanics)	4	3	3	ENGG 111, GeoE 200		TC-E
SE 301	Public Health Engineering	2	1	3	SE 204		TC-E
SE 302	SE Laws, Contracts and Ethics	3	3	0	SE 201		TC-E
ENGG 104	Engineering Economics	3	3	0	MATH 102		TC-C
AI 103	Machine Learning and Neural Networks	3	2	3			TC-F
	Total	**26**	**21**	**15**			

THIRD YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
SE 303	Water Supply Planning and Development	3	3	0	CE 205, CE 302		TC-E
SE 304	Sewerage and Urban Drainage	3	3	0	CE 308		TC-E
SE 305	Sewage and Industrial Wastewater Treatment	3	3	0	ENGG 113		TC-E
CE 202	Construction Materials and Testing	3	2	3	ENGG 111		TC-E
CE 309	Principles of Reinforced/Prestressed Concrete Design	4	3	3	CE 301		TC-E
SE 306	Solid and Hazardous Waste Engineering	3	3	0	SE 301		TC-E
ENGG 116	Methods of Research	3	3	0	AI 102, 3rd yr standing		TC-E
SE 307	Sanitary Science, Plumbing and Fire Protection as Applied to Buildings	3	2	3	CE 205, CE 302		TC-E
SE 308	GIS Application for Sanitary Engineers	1	0	3			TC-E
	Total	**26**	**22**	**12**			

THIRD YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
GEdE 101	The Entrepreneurial Mind	3	3	0			TC-F
SE 309	Environmental Planning, Laws and Impact Assessment	3	3	0			TC-E
ENGG 114	Basic Occupational Safety and Health	3	3	0			TC-E
	Total	**9**	**9**	**0**			

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FOURTH YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 117	OJT/Internship (320 hrs)	4	320 hrs	4th yr standing			TC-E
SE 401	Capstone Design Project 1	2	1	3	4th yr standing, ENGG 116		TC-E
CE 403	Construction Methods and Project Management	4	3	3	ENGG 106		TC-E
	Total	**10**	**4**	**6**			

FOURTH YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 105	Technopreneurship	3	3	0	4th yr standing		TC-C
SE 402	Capstone Design Project 2	2	0	6	SE 401, CE 403		TC-E
SE 403	Groundwater and Soil Pollution Remediation	2	2	0	CE 306		TC-E
SE 404	Environmental Engineering Laboratory	1	0	3	4th yr standing		TC-E
SE 405	Water Purification Process Design	3	3	0	SE 303		TC-E
SE 406	SE Practice with Comprehensive Examinations	2	0	6	Graduating		TC-E
	Total	**13**	**8**	**15**			

TOTAL UNITS: 196